

Reinforcement Learning With Reward Machines in Stochastic Games

Matteo Liotta and Mattia Simonutti

University of Trieste

- **Problem:** How to tackle complex tasks in multi-agent RL where reward functions are non-Markovian?
- **Proposed Approach:** Embed reward machines into the state to form an augmented product stochastic game, converting non-Markovian rewards into standard Markovian ones.
- **Goal:** Learn and converge to the optimal best-response strategy at a Nash equilibrium for each agent in a general-sum game setting.

Outline

- 1 Basic Definitions
- 2 Game Theory Setting
- 3 Unified Framework
 - Algorithmic Phase Analysis
- 4 Theoretical Foundations
- 5 Results and Applications
- 6 Conclusion

Definition: Two-Player General-Sum Stochastic Game

Modeled as the tuple $\mathcal{G} = (S, s_I, A_e, A_a, p, R_e, R_a, \gamma)$, where:

- **State Space (S):** Tracks paired substates $s = [s_e; s_a]$ for both the ego and adversarial agent.
- **Non-Markovian Rewards:** The reward functions R_e and R_a map from complete trajectory histories to real value domains $(S \times A_e \times A_a)^+ \times S \rightarrow \mathbb{R}$.
- **Action Spaces:** Each agent has its own finite set of actions A_e and A_a .

Trajectories and Rewards

- **Trajectory:** Sequence of states and actions $s_0 a_{e,0} a_{a,0} s_1 \dots s_k a_{e,k} a_{a,k} s_{k+1}$ starting from s_I .
- **Shared Observation:** Both agents observe the same environment trajectory.
- **Reward Sequence:** Each agent $i \in \{e, a\}$ receives a sequence $r_{i,1} \dots r_{i,k}$ based on the full history.
- **Discounted Cumulative Reward:** The ego agent achieves a total reward of $\sum_{t=0}^k \gamma^t r_{e,t}$, and the adversarial agent achieves $\sum_{t=0}^k \gamma^t r_{a,t}$.

Strategies and Objectives

- **Markovian Strategy:** A mapping $\pi_i : S \times A_i \rightarrow [0, 1]$ that defines the probability of selecting an action given the state.
- **Objective:** Maximize the expected discounted cumulative reward starting from an initial state s .

Expected Value Function

$$\tilde{v}_i(s, \pi_e, \pi_a) = \sum_{t=0}^{\infty} \gamma^t \mathbb{E}(r_{i,t} | \pi_e, \pi_a, s_0 = s) \quad (1)$$

where $i \in \{e, a\}$.

Reward Machines (RM)

Definition: Reward Machine

Finite state automaton designed to encode non-Markovian rewards with state transitions and output functions: $\mathcal{A} = (V, v_I, 2^{\mathcal{P}}, \mathbb{R}, \delta, \sigma)$

- **Input Alphabet ($2^{\mathcal{P}}$):** Interprets propositional logic variables $\mathcal{P}$ generated from lower-level environment updates.
- **Set of RM states (V):** Finite collection of internal tracking states, with v_I as the initial state.
- **Transition function:** Deterministic state progression
 $\delta : V \times 2^{\mathcal{P}} \rightarrow V$.
- **Output function:** Discrete scalars along active trajectory
 $\sigma : V \times 2^{\mathcal{P}} \rightarrow \mathbb{R}$.

Nash Equilibrium

Definition: Nash Equilibrium

Each agent's strategy is a best-response to the other agent's strategy, such that each agent cannot unilaterally improve its own reward by changing its own strategy.

Definition: Nash Equilibrium in Stochastic Games

For a stochastic game $\mathcal{G} = (S, s_I, A_e, A_a, R_e, R_a, \gamma, p)$, the strategies of the ego agent and the adversarial agent are at a Nash equilibrium if:

$$\begin{aligned}\tilde{v}_e(s, \pi_e^*, \pi_a^*) &\geq \tilde{v}_e(s, \pi_e, \pi_a^*) \\ \tilde{v}_a(s, \pi_e^*, \pi_a^*) &\geq \tilde{v}_a(s, \pi_e^*, \pi_a)\end{aligned}$$

hold for any π_e and π_a .

Outline

- 1 Basic Definitions
- 2 Game Theory Setting
- 3 Unified Framework
 - Algorithmic Phase Analysis
- 4 Theoretical Foundations
- 5 Results and Applications
- 6 Conclusion

Definition 4: Reward Machine Encoding

A reward machine $\mathcal{A}_i = (V_i, v_{l,i}, 2^{\mathcal{P}}, \mathbb{R}_i, \delta_i, \sigma_i)$ encodes the non-Markovian reward function R_i of agent i in the stochastic game $\mathcal{G}$ if for every trajectory and corresponding event sequence $l_0 l_1 \dots l_k$, it holds that:

$$r_{i,0} \dots r_{i,k} = \mathcal{A}_i(l_0 l_1 \dots l_k)$$

- Formalizes RMs as high-level task specifications.
- Each agent uses a separate RM ($\mathcal{A}_e, \mathcal{A}_a$) to track independent non-Markovian goals.

The Augmented State Space Framework

Labeling Function (L)

Maps environment transitions to logical propositions:

$$L : S \times A_e \times A_a \times S \rightarrow 2^{\mathcal{P}}$$

$I_t = L(s_t, a_{a,t}, a_{e,t}, s_{t+1})$ is the set of true propositions in $\mathcal{P}$ at step t .

Lemma 1: Markovian Restoration

The product stochastic game has **Markovian reward functions** that express the original non-Markovian rewards.

- Given I_t , the RM state transitions to $v_{t+1} = \delta(v_t, I_t)$ and outputs reward $r_t = \sigma(v_t, I_t)$.
- This resolves historical dependencies and enables standard Q-learning.

Definition 5: Product Stochastic Game

Given an SG $\mathcal{G}$ and RMs $\mathcal{A}_e, \mathcal{A}_a$, the product SG is:

$$\mathcal{H} = (S', s'_I, A_e, A_a, p', R'_e, R'_a, \gamma)$$

where:

- **Augmented State Space:** $S' = S \times V_e \times V_a$. An augmented state is $s' = (s, v_e, v_a)$.
- **Initial Augmented State:** $s'_I = (s_I, v_{I,e}, v_{I,a})$.
- **Transition Probabilities:** $p'(s', a_e, a_a, \bar{s}') = p(s, a_e, a_a, \bar{s})$ if $\bar{v}_i = \delta_i(v_i, L(s, a_e, a_a, \bar{s}))$ for $i \in \{e, a\}$, and 0 otherwise.
- **Markovian Reward Functions:**

$$R'_i(s', a_e, a_a, \bar{s}') = \sigma_i(v_i, L(s, a_e, a_a, \bar{s}))$$

The Stage Game Matrix

Definition: Stage Game

A two-player stage game is defined as (r_ϵ, r_α) , where r_i is agent i 's reward function R_i , over the space of joint actions:

$$r_i = \{R_i(a_\alpha, a_\epsilon) | a_\alpha \in A_\alpha, a_\epsilon \in A_\epsilon\}$$

- In QRM-SG, a stage game is formulated at each time step based on current Q-functions in augmented state space.
- The **Lemke-Howson method** is used to derive best-response strategies at a Nash equilibrium.

Nash Equilibrium & The Stage Game

Augmented Q-Function at Nash Equilibrium

Defined over the coupled augmented state space $S \times V_e \times V_a$:

$$q_i^*(s, v_e, v_a, a_e, a_a) = r_i + \gamma \sum_{s', v'} p(s', v' | s, v, a_e, a_a) \tilde{v}_i(s', v'_e, v'_a, \pi_e^*, \pi_a^*)$$

- q_{ij} represents the Q-function for agent i as learned by agent j .
- Each agent learns the Q-functions of **all agents** in the system to calculate current equilibria locally.

Outline

- 1 Basic Definitions
- 2 Game Theory Setting
- 3 Unified Framework**
 - Algorithmic Phase Analysis
- 4 Theoretical Foundations
- 5 Results and Applications
- 6 Conclusion

Actor Mechanics (Action Selection)

- Balances policy exploration using an ϵ -greedy wrapper around calculated stage-game equilibrium solutions.
- With probability $1 - \epsilon$, the model tracks the maximum probability mixed-strategy weights output by the **Lemke-Howson algorithm**.

Critic Updates (Learning Loop)

Values are updated according to standard stochastic approximation:

$$q_{ji} \leftarrow (1 - \alpha_t)q_{ji} + \alpha_t(r_{i,t} + \gamma \bar{q}_{ji}(s', v'_e, v'_a))$$

where $\bar{q}_{ji}$ is the expected value scalar at the next augmented state under best-response actions.

Process Breakdown (Lines 1–4)

Initialize $\alpha_t, \gamma, \epsilon$

$s \leftarrow \text{InitialState}(); v_{\epsilon} \leftarrow v_{I,\epsilon}; v_{\alpha} \leftarrow v_{I,\alpha}$

$q_{\epsilon i}(\cdot), q_{\alpha i}(\cdot) \leftarrow \text{InitialQfunctions}()$

Analysis:

- Each agent starts with a "dual-view" Q-table, estimating values for self and opponent to allow local equilibrium computation.
- State is defined in the product space $S \times V_{\epsilon} \times V_{\alpha}$, ensuring the Markov property is preserved.

QRM-SG Phase 2: Nash-Action Selection

Process Breakdown (Lines 7–13)

```
if  $\bar{\epsilon} < \epsilon$  then
   $a_i \leftarrow \text{ChooseActionRandomly}()$ 
else
   $\pi_{ei}, \pi_{ai} \leftarrow \text{CalculateNashEquilibrium}(q_{ei}, q_{ai})$ 
   $a_i \leftarrow \operatorname{argmax} \pi_{ij}(a|s, v_e, v_a)$ 
end if
```

Analysis:

- **Exploration:** Standard ϵ -greedy mechanism balances discovery and optimization.
- **Exploitation:** Agents solve a bimatrix stage game using the **Lemke-Howson method** to find mixed strategy equilibria.

Process Breakdown (Lines 14–18)

$$s' \leftarrow \text{ExecuteAction}(s, a_e, a_a)$$
$$l_t \leftarrow L(s, a_e, a_a, s')$$
$$v'_i \leftarrow \delta_i(v_i, l_t); r_{i,t} \leftarrow \sigma_i(v_i, l_t)$$

Analysis:

- Bridges low-level physical dynamics ($s \rightarrow s'$) with high-level automata state tracking.
- The labeling function L detects high-level environmental events (e.g., "power base reached").
- Rewards $r_{i,t}$ are generated by the RM transition, making them context and history-aware.

QRM-SG Phase 4: Q-Value Update (Critic)

Process Breakdown (Lines 19–24)

$$\pi_{ji}(\cdot | s', v') \leftarrow \text{Nash}(q_{ei}(s', v'), q_{ai}(s', v'))$$

$$\bar{q}_{ji} \leftarrow \pi_{ei} \pi_{ai} q_{ji}(s', v')$$

$$q_{ji} \leftarrow (1 - \alpha_t) q_{ji} + \alpha_t (r_{j,t} + \gamma \bar{q}_{ji})$$

Analysis:

- **Bootstrapping:** Updates current Q-values by computing the Nash equilibrium of the **future** state (s', v') .
- **Scalarization:** $\bar{q}_{ji}$ is the expected value of agent j when both follow the Nash strategy profile.
- **Coupled Update:** Agents must learn both Q-functions concurrently to maintain equilibrium local estimates.

Outline

- 1 Basic Definitions
- 2 Game Theory Setting
- 3 Unified Framework
 - Algorithmic Phase Analysis
- 4 Theoretical Foundations**
- 5 Results and Applications
- 6 Conclusion

Convergence Assumptions

Assumption 1: Exploration

Every state $s \in S$, RM states $v_e \in V_e$, $v_a \in V_a$, and actions $a_e \in A_e$, $a_a \in A_a$, are visited infinitely often when the number of episodes goes to infinity.

Assumption 2: Learning Rate

The learning rate α_t satisfies the following conditions for all t :

1. $0 < \alpha_t < 1$, $\sum_t \alpha_t(s, v_e, v_a, a_e, a_a) = \infty$, $\sum_t [\alpha_t(s, v_e, v_a, a_e, a_a)]^2 < \infty$, and the latter two hold uniformly and with probability 1.
2. $\alpha_t(s, v_e, v_a, a_e, a_a) = 0$ if $(s, v_e, v_a, a_e, a_a) \neq (s^t, v_e^t, v_a^t, a_e^t, a_a^t)$.

Assumption 3: Stage Game Structure

One of the following conditions holds during learning:

1. Every stage game $(q_{ei}(s, v_e, v_a), q_{ai}(s, v_e, v_a))$, $i \in \{e, a\}$, for all t, s, v_e , and v_a , has a **global optimal point** $(\tilde{\pi}_{ei}, \tilde{\pi}_{ai})$ (i.e., $\tilde{\pi}_{ji}q_{ji} \geq \tilde{\pi}'_{ji}q_{ji}$ for any $\tilde{\pi}'_{ji}$, $j \in \{e, a\}$) and agents' rewards in this equilibrium are used to update their Q-functions.
2. Every stage game $(q_{ei}(s, v_e, v_a), q_{ai}(s, v_e, v_a))$, $i \in \{e, a\}$, for all t, s, v_e , and v_a , has a **saddle point** $(\tilde{\pi}_{ei}, \tilde{\pi}_{ai})$ (i.e., $\tilde{\pi}_{ji}\tilde{\pi}_{-ji}q_{ji} \geq \tilde{\pi}'_{ji}\tilde{\pi}_{-ji}q_{ji}$ for any $\tilde{\pi}'_{ji}$, $\tilde{\pi}_{ji}\tilde{\pi}_{-ji}q_{ji} \leq \tilde{\pi}_{ji}\tilde{\pi}'_{-ji}q_{ji}$ for any $\tilde{\pi}'_{-ji}$, $j \in \{e, a\}$), and agents' rewards in this equilibrium are used to update their Q-functions.

Theorem 1: Convergence Guarantee

Theorem 1

Under Assumptions 1, 2 and 3, the sequence $q_{it} = (q_{ei}^t, q_{ai}^t)$ at time t , $i \in \{e, a\}$ updated by:

$$q_{ji}(s, v, a) = (1 - \alpha_t)q_{ji}(s, v, a) + \alpha_t(r_{j,t} + \gamma \pi_{ei}(\cdot | s', v') \pi_{ai}(\cdot | s', v') q_{ji}(s', v'))$$

for $j \in \{e, a\}$, where $(\pi_{ei}(\cdot | s', v'), \pi_{ai}(\cdot | s', v'))$ is the appropriate type of Nash equilibrium solution for the stage game $(q_{ei}^t(s', v'), q_{ai}^t(s', v'))$, converges to the Q-functions at a Nash equilibrium (q_{ei}^*, q_{ai}^*) .

Algorithm 1: QRM-SG Full Pseudocode

```
1: Initialize  $q_{ei}$  and  $q_{ai}$  for all  $i \in \{e, a\}$ 
2: for episode  $l = 1, 2, \dots$  do
3:    $s \leftarrow s_l, v_e \leftarrow v_{l,e}, v_a \leftarrow v_{l,a}$ 
4:   for  $t = 0$  to  $eplength$  do
5:     Choose action  $a_i$  using  $\epsilon$ -greedy policy and Lemke-Howson
6:     Execute actions, observe  $s'$ , and high-level event
        $l_t = L(s, a_e, a_a, s')$ 
7:     Update RM states:  $v'_i \leftarrow \delta_i(v_i, l_t)$  and reward  $r_{i,t} \leftarrow \sigma_i(v_i, l_t)$ 
8:     Compute Nash strategy  $\pi_{ji}(\cdot | s', v')$  at next augmented state
9:     Update Q-functions:
10:     $q_{ji} \leftarrow (1 - \alpha_t)q_{ji} + \alpha_t(r_{j,t} + \gamma \sum \pi \pi q'_{ji})$ 
11:     $s \leftarrow s', v_e \leftarrow v'_e, v_a \leftarrow v'_a$ 
12:   end for
13: end for
```

Algorithm Analysis and Convergence Properties

- **Off-Policy Learning:** RMs enable decomposition of complex RL problems into structured subproblems for efficient learning.
- **Coupled Learning:** The learning processes of agents are coupled because the high-level events depend on joint actions.
- **Decentralized View:** Each agent maintains a "centralized" perspective by learning everyone's Q-functions to compute equilibria locally.

Mathematical Foundation: Convergence Proof

The proof relies on viewing the update as a stochastic approximation of a mapping F^t :

Key Lemmas from Supplementary Materials

- **Lemma 2:** Establishes general convergence for iterations $q^{t+1} = (1 - \alpha_t)q^t + \alpha_t[F^t q^t]$.
- **Lemma 4:** Shows that Q-functions at a Nash equilibrium (q^*) are fixed points of F^t , i.e., $q^* = \mathbb{E}[F^t q^*]$.
- **Lemma 7:** Proves F^t is a **contraction mapping** under Assumption 3:

$$\|F^t q - F^t \dot{q}\| \leq \gamma \|q - \dot{q}\|$$

Outline

- 1 Basic Definitions
- 2 Game Theory Setting
- 3 Unified Framework
 - Algorithmic Phase Analysis
- 4 Theoretical Foundations
- 5 Results and Applications
- 6 Conclusion

- **Testing Environment:** Evaluated on 6×6 (and 12×12) competitive, sparse-reward grid world variations of Pac-Man.
- **Comparative Baselines:**
 - Nash-Q: Pure spatial state-action tracking.
 - Nash-QAS: Incorporates an unstructured history status vector.
 - MADDPG-SG & MADDPG-AS: Deep multi-agent deterministic policy gradient variants.

Case Study Empirical Analysis

- **Case Study I (Sequential Ordering):** QRM-SG converges to equilibrium by $\approx 7,500$ episodes, whereas unaugmented baselines fail entirely.
- **Case Study II & III (Energy Loops):** Explores complex recharge loops with randomized origin points. QRM-SG reaches stability within $\approx 1,000$ to $1,500$ runs.
- **Scalability:** Successfully finds the Nash equilibrium on an expanded 12×12 map space by episode 21,000.

Outline

- 1 Basic Definitions
- 2 Game Theory Setting
- 3 Unified Framework
 - Algorithmic Phase Analysis
- 4 Theoretical Foundations
- 5 Results and Applications
- 6 Conclusion**

Summary & Future Outlook

- **Core Contribution:** Bridges reward machines with MARL to model non-Markovian constraints in stochastic games.
- **Robust Architecture:** Augmented state transformations allow tabular actors to navigate complex environments successfully.
- **Future Directions:** Dynamically learning unknown reward machine configurations simultaneously during policy calculation; extensions to general-sum games with more agents.